

M.AI.ESTRO VR: system audit

Finalized 4 October 2026 · Graham Pellegrini · University of Malta

System Demo

Full recording stitch

[Open full recording](#)

Clock orbit: capture and replay

RGB / SAM2 on the left · Unity ghost replay on the right · Effective 3× speed (video encoded at 1.5×, player at 2×). [Open orbit comparison](#)

Clock task: capture and replay

RGB / SAM2 on the left · Unity ghost replay on the right. Task replay uses the latest orbit mesh under the temporary demo policy. [Open task comparison](#)

Pipeline Diagram

The pipeline PNG shows the flow from headset capture to stored object evidence, with separate task and orbit branches. Open the full-resolution image to inspect individual labels; the PDF gives the complete diagram its own landscape A3 page.

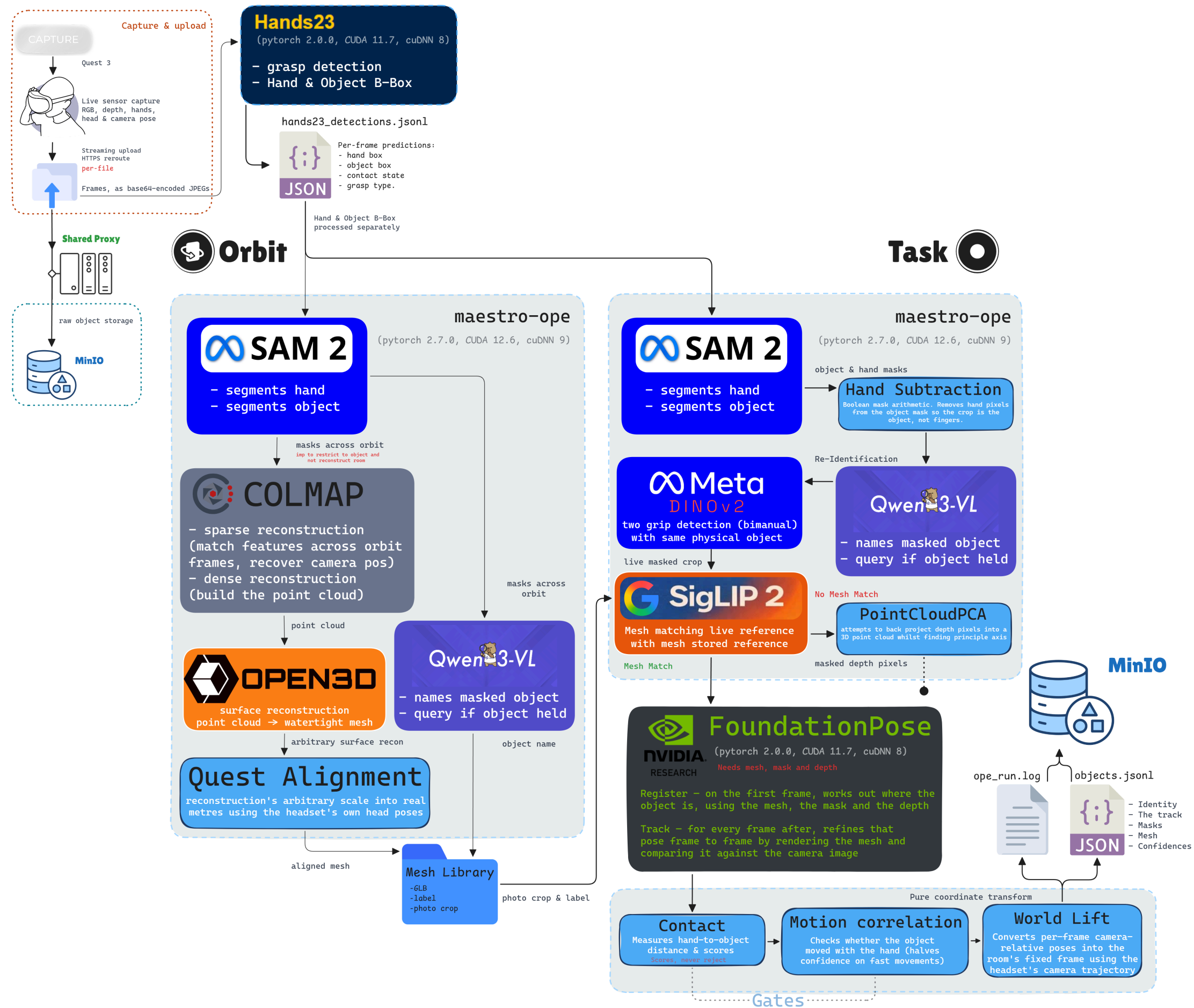


Figure 1. Current Maestros VR pipeline used for the audit discussion.

Pipeline Modules

Each node has a narrow responsibility. The audit question is whether the handoff between nodes is reliable enough for replay.

1

Quest capture

Records RGB, depth, headset pose and hand pose for task and orbit sessions.

[Quest sensor capture](#)

2

Ingestion and MinIO

Accepts uploads, stores raw inputs and exposes session artifacts for worker processing.

[MinIO object storage](#)

3

Hands23 hand detection

Provides hand/object contact evidence and candidate boxes before object processing begins.

[Hands23 code and explanation](#)

4

SAM2 segmentation

Turns candidate evidence into masks for orbit reconstruction and task-frame object tracking.

[SAM2 overview and code](#)

5a

PointCloudPCA

Provides model-free task pose estimates when a trusted mesh is not available.

[Project pose measurements](#)

5b

FoundationPose with mesh

Runs when a suitable mesh, mask and depth are available; this is the stronger route for mesh-backed replay.

[FoundationPose overview and code](#)

6

Qwen labels

Supplies an object label from the crop; useful, but still dependent on mask quality.

[Qwen3-VL-4B-Instruct model](#)

7

DINOv2 and SigLIP

Support visual identity checks. The current audit keeps DINOv2 and SigLIP rather than adopting DINOv3.

[DINOv2 ViT-S/14 model](#) · [SigLIP base model](#)

8

COLMAP mesh reconstruction

Reconstructs object meshes from orbit capture frames for canonical mesh-backed replay.

[COLMAP tutorial and code](#)

9

Headset replay

Consumes stored objects, tracks, meshes and masks so captured objects can be inspected in VR.

[Unity XR Interaction Toolkit](#)

System Status

Confirmed steps come first, followed by demo patches, partial fixes and disabled work, then outcomes still to confirm. Open up to two cards at once. The side connections show where a working step still depends on unfinished follow-up.

Hands23: the starting point

A detected grip supplies the hand and object boxes that seed the rest of the pipeline.

Module 3 **Confirmed**

The live worker checks the separate Hands23 service and saves its detections before object processing. Missing detections now raise an error, instead of being mistaken for a recording with no object. A brief deliberate grip seeds the orbit route; task recordings use contact evidence to start tracking.

These boxes changed the architecture: SAM2 can now isolate both the hand and the object, so a separate SAM3 hand detector is no longer needed. A box is the starting region, not a finished mask or mesh.

[Hands23 code and explanation](#)

Grip-first orbit mesh generation

The reconstruction route produces real-scale meshes when a grip is detected and the orbit has usable visual detail.

Modules 3, 4, 8 **Confirmed**

Grip first, then slowly circle the object for the rest of the clip. Hands23 identifies the starting object; SAM2 follows its outline. Masked COLMAP matches visible features across the images and reconstructs a 3D point cloud. Open3D builds the surface, and Quest head poses set its scale in metres.

Candle and clock reconstructions are measured successes. This confirms the reconstruction method; it does not establish unattended upload-to-mesh automation for every session. The latest clock reconstruction was run and registered manually.

Route	Input and purpose	Project finding
Hunyuan3 D-Omni	One reference image; generates plausible hidden surfaces	Poor remote-control result despite a clean photograph.
TRELLIS. 2	One reference image; generates a textured mesh	Cleaner remote-control result; unseen surfaces and orientation still need checking.
Fuse3D	Several real views fused through a generative model	Strong front, thinner sides in the same comparison.
COLMAP + Open3D	Matches real image features across an orbit; reconstructs points, then a surface	Measured metric candle and clock meshes. Low-texture objects can still fail.

The earlier comparison is a small project trial, not a general model ranking. COLMAP is the measured orbit route, but the bottle and rolling pin failed at feature matching. The bowl had healthy pairwise matches and failed later in multi-view consistency.

Follow-ups: [mesh selection and replay](#) · [orientation and interaction](#).

[COLMAP tutorial and code](#)

FoundationPose with a trusted mesh

The mesh-backed service has run successfully; reliable tracking through hand occlusion still needs care.

Module 5b **Confirmed**

FoundationPose was already implemented. The recorded activation blockers were the wrong GPU being almost full from another job, and a mesh loader expecting OBJ while registered assets used

GLB. Moving the service to a GPU with room and supplying the expected mesh enabled real registration and more than 100 successful tracking calls.

The bottle trial still had 9 large rotation jumps in 112 transitions, mostly flip-and-recover pairs. Hand occlusion was a likely cause, not a proven diagnosis. A working container does not prove perfect tracking.

Default live task processing remains model-free while mesh matching is disabled. Borrowing a mesh for rendering does not rerun FoundationPose.

Without a trusted mesh, the remaining fallback issue is [PointCloudPCA](#).

[FoundationPose overview and code](#)

SAM3 removal: SAM2 is sufficient for the current prompts

Hands23 supplies the boxes; SAM2 creates the object and hand masks.

Modules 3, 4 Confirmed

The earlier reference-image mesh route needed hand pixels removed before generating a mesh. SAM3 was also used for an object-presence check. It was removed after the Hands23-guided SAM2 route was evaluated; this was more than removing a duplicate model.

Model	Plain-language role	Why this pipeline changed
SAM2	Select a region using a point or box, then follow its pixels through video.	Hands23 supplies an object box and a hand box. SAM2 makes separate object and hand masks; hand pixels are subtracted.
SAM3	Find and segment concepts using text or example prompts.	Previously used for hand removal and an object-presence check. Its hand mask failed on a real bottle recording; the presence score did not reliably distinguish a real grasp from a fake one.

SAM2 runs with two different prompts, producing two masks. Qwen performs the remaining held-object check. This keeps the pipeline simpler without claiming SAM3 is inferior for every task.

[SAM2 overview](#) · [SAM3 overview](#)

Episode naming endpoint

Names are built from existing object labels and capture dates, then served to Unity by session UUID.

`{object_key}_{counter}_{YYMMDD}`

The server uses existing Qwen3-VL labels and, when available, canonical category or label data. It normalises supported terms, rather than asking another model to invent a task title. The counter starts at 00 for each object-key/date pair and grows beyond 99. The capture date prefers device-local time.

Outcome	Example display name	Meaning
Waiting / processing	<code>unidentified_object_00_261005</code>	Pending; the object is not yet identified.
Finished, no object	<code>unidentified_object_00_261005</code>	Unidentified; distinct processing state even though the name has the same form.
One recognised object	<code>clock_00_261005</code>	A supported token from the existing label or canonical record.
Two different objects	<code>bottle.clock_00_261005</code>	Distinct tokens sorted alphabetically and joined with a dot.
Repeated object type	<code>clock_x2_00_261005</code>	Two tracked records with the same token.
Three different types	<code>bottle.candle.clock_00_261005</code>	Up to three distinct tokens remain explicit.
Four or more types	<code>multiobj4_00_261005</code>	Number counts distinct types, rather than total objects.
Unsupported label	<code>unidentified_object_00_261005</code>	A detected object can still have an unsupported name; check the state and label.
Device legacy title	<code>untitled</code>	Old device title, not a generated server naming outcome.

`GET /sessions/{uuid}/metadata` includes the display name, processing state, revision, timestamp and grasp result. Pending and finished-with-no-object remain distinct. Device folders are not renamed.

Mesh selection and headset replay

Meshes can render in replay; the current task selection policy is a temporary shortcut.

Modules 7, 9

Fixed for demo

- **Rendering:** demo sessions had meshes assigned manually. The clock bounding-box mismatch was corrected so Unity's shape check accepts the mesh.
- **Orbit ownership:** orbit replay uses its own registered reconstruction rather than another session's mesh.
- **Task shortcut:** every task object is currently served the latest mesh-bearing orbit registration, without checking object identity. Source orbit dimensions accompany the mesh; task tracks remain unchanged.
- **Matching disabled:** SigLIP retrieval achieved 4/11 correct historical decisions; correct and incorrect scores overlap. Automatic matching remains switched off.
- **Still needed:** validated identity and mesh-quality selection, mesh-backed pose estimation where appropriate, and headset checks of the latest API policy.

This patch joins display and storage; it does not make the borrowed mesh the right object or improve the recorded poses.

PointCloudPCA: pose fallback without a mesh

A partial view or noisy depth can make the estimated object position jump.

Module 5a

Partially fixed

This route estimates the centre and orientation of the visible depth points. If the visible patch changes, the estimated object can suddenly move even when the real object does not. The full object shape is unknown.

- | Slow candle: maximum frame-to-frame jump 0.153 m.
- | Fast clock: maximum frame-to-frame jump 1.053 m.
- | Table-contact Stanley cup: maximum jump 0.738 m.

These are recorded pre-guard measurements, not new results after the fix. During a confirmed grip, the deployed guard drops estimates more than 0.5 m from every available wrist/palm. Missing hand landmarks cannot be checked, and unheld objects are not covered by this rule.

Next: measure post-guard natural-motion performance and add justified motion, table-contact or temporal constraints. Robust fallback replay remains unconfirmed.

Automatic mesh matching

The identity matcher stays off while false assignments remain unresolved.

Module 7 **Disabled**

The full mesh-selection situation is collected in [Mesh selection and headset replay](#). This disabled branch is why normal live tasks do not automatically reach FoundationPose.

Replay orientation and interaction

Unity-side facing, placement, size and grabbing still need refinement.

Module 9 **Not confirmed**

- Align ghost replay with the viewer's current facing direction at spawn.
- Correct clock orbit mesh orientation.
- Define placement when new recording stamps are added, and check their sizing.
- Support both direct grabbing and Quest-style distance pointers.

These are reported follow-ups, not verified fixes. The latest mesh policy does not resolve them.

Dedicated Maestro server endpoint

The IT Services request is submitted; there has been no follow-up response.

Module 2 **Not confirmed**

Maestro currently uses the shared university proxy. A dedicated endpoint is pending for EEAI 2026 and the presentation of the encapsulated edge-AI system. A dedicated endpoint is the intended infrastructure change, but successful operation after migration is not yet established.

After IT supplies it, validate headset URLs, HTTPS, uploads, metadata and mesh downloads together; changing the endpoint may expose integration problems.

DINOv3 encoder consolidation

A single encoder could replace DINOv2 and SigLIP, but the corrected results do not justify adoption.

Module 7 **Not confirmed**

The aim is to simplify visual comparison and reduce model weight. DINOv3-S has 21.60M parameters and B 85.66M, versus 225.21M for the existing pair. The proposed replacement is not deployed.

Method	Same object across hands, at 0.82	Correct mesh decisions: historical 11 / all 15
Current DINOv2 + SigLIP	DINOv2: 3/4	SigLIP: 4/11 · 7/15
DINOv3-S	2/4	5/11 · 8/15
DINOv3-B	2/4	5/11 · 7/15
DINOv3-B foreground comparison	Not measured as a separate re-ID method	10/11 · 13/15; still two absent objects assigned incorrectly

The earlier test compared reused/self crops rather than independent hand-to-hand views, and some source sessions were empty or had wrong masks. That proxy could not establish real re-identification. The corrected evaluation restored the data and used independent crops; it is complete and supersedes that first test.

Foreground matching ranks known objects better, but still wrongly assigns an absent pen and remote. Next research needs independent held-out recordings, absent-object checks and reliable rejection of unknown objects; any adopted replacement must also preserve the identity pipeline's spatial/time checks. Keep DINOv2 and SigLIP, and leave mesh matching disabled meanwhile.

[Corrected evaluation and methods](#)

[DINOv3 overview and code](#)

Possible feedback and design decisions

- **Naming convention:** should users keep the compact machine name, see a friendlier title, or add their own task title? Current names describe object records, not the action performed.
- **Offline Wi-Fi:** distinguish connected, unstable and fully offline states with an explicit icon/message. Agree how queued uploads and unavailable metadata should appear; client behaviour still needs validation.

Technical Detail

[Full technical audit and measurements](#)